

Notes: Adams & Ferreira (JFE, 2009)

Women in the Boardroom and Their Impact on Governance and Performance

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1 Big picture

- **Question.** Does gender diversity on corporate boards change board behaviour, governance outcomes, and firm performance?
- **Core setting.** The paper studies an unbalanced panel of U.S. firms covered by IRRC director data and ExecuComp from 1996–2003. The unit of observation changes across tests: director-year for attendance and committees, firm-year for compensation, CEO turnover, and performance.
- **Main board-input result.** Female directors have better attendance records than male directors. Male directors also have fewer attendance problems when boards are more gender diverse.
- **Committee result.** Female directors are more likely to serve on monitoring-oriented committees such as audit, nominating, and corporate governance committees, but less likely to serve on compensation committees.
- **Governance result.** Boards with more women appear to monitor more intensely. CEO turnover is more sensitive to poor stock performance when the fraction of female directors is higher.
- **Director-pay result.** Directors on gender-diverse boards receive more equity-based compensation, consistent with stronger incentive alignment for directors.
- **CEO-pay result.** Gender diversity does not have a strong or robust relation with CEO pay levels or CEO incentive pay.
- **Performance result.** Naive correlations between female board representation and firm value can be positive, but after firm fixed effects, IV, and dynamic panel controls, the average effect of gender diversity on performance is negative or weak.
- **Governance interaction.** The negative performance effect is concentrated in firms with strong shareholder rights. Gender diversity appears beneficial in firms with weak shareholder rights, where extra monitoring is more valuable.
- **Policy implication.** Female directors have real governance effects, but mandatory quotas can reduce value for firms that already have strong governance because extra monitoring may become excessive.

2 Data and measurement

2.1 IRRC director data and ExecuComp sample

- The core sample covers Standard & Poor’s 500, MidCap, and SmallCap firms appearing in IRRC director data and ExecuComp.
- The sample period is 1996–2003.
- Table 1 reports 86,714 director-level observations and 8,253 firm-year observations with the main variables.
- IRRC provides director identity, gender, independence, committee assignments, attendance disclosures, and board characteristics.
- ExecuComp provides CEO compensation, director compensation, and some financial variables.
- Compustat and CRSP provide accounting variables, stock returns, SIC codes, and return volatility.

Interpretation: The data allow the authors to observe actual board inputs, not only board composition. This is important because the paper’s mechanism is about monitoring effort and governance activity.

2.2 Key gender-diversity variables

- **Female Dummy:** equals one if a director is female.
- **Firm Has Female Directors:** equals one if a firm has at least one female director in a given year.
- **Firm Has Only One Female Director:** equals one if a firm has exactly one female director.
- **Fraction Female Directors:** number of female directors divided by board size.
- The mean fraction of female directors in the main firm-year sample is 0.085.

Interpretation: The paper distinguishes individual gender effects from board-level gender diversity. This matters because female directors may behave differently, and male directors may also change behaviour on more gender-diverse boards.

2.3 Director attendance and committee variables

- **Attendance Problem:** equals one if the proxy reports that a director attended less than 75% of board and committee meetings.
- Committee variables equal one if a director sits on a particular committee.
- The paper focuses on audit, compensation, nominating, and corporate governance committees.

Interpretation: Attendance is a direct measure of director effort. Committee membership identifies whether women are assigned to monitoring-sensitive functions.

2.4 Governance and performance variables

- **CEO Turnover:** a firm-year indicator for CEO replacement.
- **Stock Performance:** the firm's raw stock return for the year net of the CRSP value-weighted index, compounded continuously.
- **Total Director Compensation:** retainer, meeting fees, and value of stock-based director compensation.
- **Fraction Equity Pay:** equity-based director compensation divided by total director compensation.
- **Total CEO Compensation:** ExecuComp's total CEO pay measure.
- **Fraction CEO Incentive Pay:** $1 - (Salary + Bonus) / Total\ CEO\ Compensation$.
- **Tobin's q:** market value divided by book assets.
- **ROA:** net income before extraordinary items and discontinued operations divided by book assets.
- **G-index:** Gompers, Ishii, and Metrick governance index, measuring antitakeover provisions and weak shareholder rights.

Interpretation: These outcomes separate governance mechanisms from performance consequences. The paper first asks whether women change board behaviour, then asks whether the resulting monitoring is valuable.

3 Models and empirical specifications

3.1 Director-level attendance regressions

The baseline director-level specification is:

$$AttendanceProblem_{i,j,t} = \alpha + \beta_1 Female_{i,j,t} + \beta_2 FractionFemale_{j,t} + \Gamma X_{i,j,t} + \delta_t + \delta_{industry} + \varepsilon_{i,j,t}. \quad (1)$$

Firm fixed-effect variants replace industry effects with firm effects:

$$AttendanceProblem_{i,j,t} = \alpha_j + \beta_1 Female_{i,j,t} + \beta_2 FractionFemale_{j,t} + \Gamma X_{i,j,t} + \delta_t + \varepsilon_{i,j,t}. \quad (2)$$

Interpretation: $\beta_1 < 0$ means female directors personally have fewer attendance problems. $\beta_2 < 0$ means all directors, especially men in the men-only sample, behave better on more gender-diverse boards.

3.2 New male director robustness

To check whether gender diversity proxies for board refreshment, the paper estimates attendance regressions including:

$$FractionNewMen_{j,t}, \quad LaggedFractionNewMen_{j,t-1}, \quad NewDirector_{i,j,t}. \quad (3)$$

Interpretation: This test separates gender diversity from the possibility that newly appointed male directors, rather than female directors, explain lower attendance problems.

3.3 Committee assignment regressions

For committee membership outcomes:

$$CommitteeMember^k_{i,j,t} = \alpha + \beta Female_{i,j,t} + \Gamma X_{i,j,t} + \delta_t + \delta_{industry} + \varepsilon_{i,j,t}, \quad (4)$$

where k indexes any committee, audit, compensation, nominating, and corporate governance committees.

Interpretation: A positive coefficient on Female Dummy means women are more likely to be assigned to the committee after controlling for experience, age, tenure, independence, and firm characteristics.

3.4 CEO turnover-performance sensitivity

The CEO turnover test is:

$$CEOTurnover_{j,t} = \alpha_j + \beta_1 FractionFemale_{j,t} + \beta_2 StockPerformance_{j,t} + \beta_3 (FractionFemale_{j,t} \times StockPerformance_{j,t}) + \Gamma X_{j,t} + \delta_t + \varepsilon_{j,t}. \quad (5)$$

Interpretation: The key coefficient is β_3 . A negative coefficient means CEO turnover becomes more sensitive to poor stock performance when the board has more women.

3.5 Director compensation and CEO compensation

For director pay:

$$Y_{j,t} = \alpha_j + \beta FractionFemale_{j,t} + \Gamma X_{j,t} + \delta_t + \varepsilon_{j,t}, \quad (6)$$

where Y is either total director compensation or fraction equity pay.

For CEO pay:

$$Y_{j,t}^{CEO} = \alpha_j + \beta FractionFemale_{j,t} + \Gamma X_{j,t} + \delta_t + \varepsilon_{j,t}. \quad (7)$$

Interpretation: Director pay tests show whether gender-diverse boards change director incentives. CEO pay tests show whether gender diversity affects executive pay design.

3.6 Performance regressions

Performance specifications use Tobin's q and ROA:

$$Performance_{j,t} = \alpha_j + \beta FractionFemale_{j,t} + \Gamma X_{j,t} + \delta_t + \varepsilon_{j,t}. \quad (8)$$

The paper estimates ordinary least squares, firm fixed effects, IV with fixed effects, and Arellano–Bond dynamic panel specifications.

Interpretation: The performance regressions are designed to address omitted variables and reverse causality. The comparison between OLS and fixed-effects/IV estimates is central to the paper.

3.7 Governance interaction regressions

To test whether diversity is more valuable in weak-governance firms, the paper estimates:

$$Performance_{j,t} = \alpha_j + \beta_1 FractionFemale_{j,t} + \beta_2 Gindex_{j,t} + \beta_3 (Gindex_{j,t} \times FractionFemale_{j,t}) + \Gamma X_{j,t} + \delta_t + \varepsilon_{j,t}. \quad (9)$$

Interpretation: Because a higher G-index means weaker shareholder rights, a positive interaction suggests that gender diversity is more valuable when baseline governance is weak.

4 Identification and empirical design

4.1 Main challenge: endogenous board composition

- Firms do not randomly appoint women to boards.
- Firms with female directors are larger, have larger boards, different q, different ROA, and different volatility.
- Board diversity may be selected by firms that already differ in unobserved governance quality or performance.

Interpretation: Simple correlations cannot be interpreted causally. Table 2 makes this problem explicit by comparing firms with and without women directors.

4.2 How the paper addresses endogeneity

- Director-level controls account for age, tenure, compensation, board meetings, board size, independence, and other directorships.
- Firm fixed effects remove time-invariant firm heterogeneity.
- Industry and year fixed effects absorb common industry and macro trends.
- IV specifications use male directors' board connections to female directors as an instrument for gender diversity.
- Arellano–Bond dynamic panel models include lagged performance and address persistence.

Interpretation: The paper's strongest performance conclusions rely on specifications that remove firm-level heterogeneity and performance persistence.

4.3 Why governance inputs are easier to interpret

- Attendance and committee assignments are observed at the director level.
- The female dummy can be compared across directors in the same types of firms.
- Male-only attendance regressions show spillovers from women on boards to male director behaviour.

Interpretation: The evidence that women change board inputs is more direct than the performance evidence. The paper is therefore strongest on mechanisms and more cautious on causal performance claims.

4.4 Policy relevance

- Gender quotas are motivated by the belief that women improve governance.
- The paper confirms that female directors increase monitoring intensity.
- But more monitoring is not always value-increasing.
- The policy conclusion is conditional: diversity can help weakly governed firms but harm well-governed firms.

Interpretation: The paper argues against one-size-fits-all quota policies because the value of monitoring depends on the firm's existing governance environment.

5 Tables and figures

5.1 Table 1: Summary statistics

Read:

- The main sample includes 8,253 firm-year observations and 86,714 director-level observations.
- Mean board size is 9.38 directors, and mean fraction independent directors is 0.63.
- 61% of firm-years have at least one female director, but the mean fraction of female directors is only 0.085.
- The mean female director dummy at the director level is 0.093.
- The attendance problem rate is low, 2.4%, but meaningful because it reflects serious attendance failures.

Economic message: The summary statistics show that female directors are present in many firms but remain a small fraction of board seats. This makes it possible to test whether even limited female representation changes board behaviour.

Table 1

Summary statistics.

The sample consists of an unbalanced panel of 86,714 director level observations from 1,939 firms for the period 1996–2003, which were both in the Investor Responsibility Research Center (IRRC) Director Data and ExecuComp. The IRRC Director data consist of director-level data for Standard & Poor's 500, S&P MidCap, and S&P SmallCap firms. Data on board meetings, director compensation, and most financial data are from ExecuComp. We obtain additional financial information (e.g., standard industrial classification codes, business segment data, and stock returns) from Compustat and the Center for Research in Security Prices. Tobin's q is the ratio of the firm's market value to its book value of assets. Market value is book assets minus book equity plus market value of equity. ROA is net income before extraordinary items and discontinued operations divided by book assets. Volatility is the standard deviation of previous 60-month stock returns. Stock Performance is the firm's raw return for the year net of the CRSP value-weighted index, both compounded continuously. The IRRC classifies directors as independent if they have no business relation with the firm, are not related or interlocked with management, and are not current or former employees. Total Director Compensation is the sum of the annual retainer, the number of board meetings times Board Meeting Fee, and the value of all stock-based compensation. Options were priced following the method in ExecuComp. We used the stock price at the end of the month of the firm's annual meeting for the exercise price of the options, as well as to value stock grants. Fraction CEO Incentive Pay is $1 - (\text{Salary} + \text{Bonus}) / \text{Total CEO Compensation}$. Total CEO Compensation is from ExecuComp and is the sum of Salary, Bonus, Other Annual, Total Value of Restricted Stock Granted, Total Value of Stock Options Granted (using Black and Scholes), Long-Term Incentive Payouts, and All Other Total. All compensation numbers have been converted to 2003 dollars using the CPI-U (consumer price index—all urban consumers). Firm Has Female Director is a dummy variable that is one if the firm has female directors in a given year. Firm Has Only One Female Director is a dummy variable that is one if the firm only has one female director. Attendance Problem is a dummy variable that is equal to one in a given fiscal year if a firm disclosed in its proxy statement that a director attended less than 75% of the meetings he was supposed to during that year. Tenure is equal to the number of years the director has served on the board. Retired Dummy is equal to one if the proxy indicated that the director retired from his primary occupation. Committee variables are dummy variables equal to one if a director sits on a committee of a given type. We show summary statistics for all observations for which Fraction Female Directors and the firm's and board level controls we use in all tables except Tables 5 and 6 are not missing.

Variable	Number of observations	Mean	Standard deviation	Min	Max
<i>Firm characteristic</i>					
Sales (millions)	8,253	4,348	9,955	0.32	168,919
Log(Sales)	8,253	7.26	1.49	−1.15	12.04
# Business Segments	8,253	5.90	4.57	1	32
Tobin's q	8,253	2.09	2.07	0.48	77.64
ROA	8,253	3.19	16.38	−57.85	59.59
Volatility	8,253	0.42	0.20	0.10	1.89
Stock Performance	6,901	−0.038	0.469	−3.77	2.01
<i>Board characteristic</i>					
Board Size	8,253	9.38	2.68	3	39
Fraction Independent Directors	8,253	0.63	0.18	0	0.94
Firm Has Female Directors	8,253	0.63	0.49	0	1
Firm Has Only One Female Director	8,253	0.40	0.49	0	1
Fraction Female Directors	8,253	0.085	0.083	0	0.5
Total Director Compensation (thousands)	8,253	95.89	198.9	0	7,973
Fraction Equity Pay	8,253	0.395	0.36	0	1
Meeting Fee (thousands)	8,253	1.03	0.85	0	7.7
Total CEO Compensation (thousands)	8,114	4,829	13,227	0	580,641
Fraction CEO Incentive Pay	8,098	0.52	0.27	0	1
<i>Director characteristic</i>					
Female Dummy	86,714	0.093	0.29	0	1
Attendance Problem	86,714	0.024	0.153	0	1
# Other Directorships	86,714	0.92	1.31	0	10
Tenure	86,714	9.7	8.1	0	63
Age	86,714	58.9	8.6	25	98
Retired Dummy	86,714	0.18	0.39	0	1
Committee Member	68,235	0.69	0.46	0	1
Audit Committee Member	68,235	0.39	0.49	0	1
Compensation Committee Member	68,235	0.37	0.48	0	1
Nomination Committee Member	68,235	0.29	0.45	0	1
Corporate Governance Committee Member	68,235	0.17	0.38	0	1

5.2 Table 2: Firms with and without female directors

Read:

- Firms with female directors are larger: mean log sales is 7.78 versus 6.47.
- They have more business segments and larger boards.
- They have lower Tobin's q but higher ROA.
- They have lower stock-return volatility.

Economic message: Firms with and without female directors differ substantially. This table motivates the paper's extensive use of fixed effects, controls, IV, and dynamic panel methods.

Table 2

Comparisons of firms with female directors to those without.

This table shows comparisons of means of firm-level characteristics for firm-years in which firms have female directors to firm-years without female directors, for the sample of complete data resulting from the intersection of the Investor Responsibility Research Center Director data, ExecuComp, Compustat, and the Center for Research in Security Prices. The number of observations is 8,253. Tobin's q is the ratio of the firm's market value to its book value of assets. Market value is book assets minus book equity plus market value of equity. ROA is net income before extraordinary items and discontinued operations divided by book assets. Volatility is the standard deviation of previous 60 month stock returns. *** indicates significance at the 1% level.

Firm characteristic	Mean for firm-years with female directors, n = 5,006	Mean for firm-years without female directors, n = 3,247	Difference
Log(Sales)	7.78	6.47	1.309***
# Business Segments	6.58	4.87	1.715***
Tobin's q	2.03	2.19	−0.165***
ROA	4.52	1.16	3.36***
Volatility	0.37	0.50	−0.126***
Board Size	10.37	7.85	2.514***

5.3 Tables 3 and 4: Attendance problems and gender diversity

Read:

- Table 3 shows that female directors have significantly fewer attendance problems than male directors.
- The Female Dummy coefficient is negative and statistically significant in both industry-effect and firm-effect specifications.
- The fraction of female directors is also negatively related to attendance problems.
- Men-only specifications show that male directors have fewer attendance problems when the board has more women.
- Table 4 shows the result is not explained by newly appointed male directors or board refreshment.

Economic message: Women appear to increase board effort both directly, through their own attendance, and indirectly, by changing the behaviour of male directors.

5.4 Table 5: Assignment of women to committees

Read:

- Female directors are more likely to be on committees overall.
- Women are more likely to sit on audit, nominating, and corporate governance committees.
- Women are less likely to sit on compensation committees.
- These patterns survive controls for attendance history, other directorships, meetings, board size, independence, tenure, age, and firm characteristics.

Economic message: Female directors are not passive tokens. They are assigned to important monitoring committees, especially those related to oversight and governance structure.

5.5 Table 6: CEO turnover and gender diversity

Read:

- CEO turnover is negatively related to stock performance.
- The interaction between Fraction Female Directors and Stock Performance is negative and statistically significant.
- This means poor stock performance is more likely to trigger CEO turnover when boards are more gender diverse.
- The comparable interaction between board independence and stock performance is not robustly significant.

Economic message: Gender diversity strengthens CEO accountability. This is one of the paper's clearest pieces of evidence that diverse boards monitor more intensely.

5.6 Tables 7 and 8: Director pay and CEO pay

Read:

- Table 7 shows that more gender-diverse boards have higher director compensation and higher equity-based director pay.

Table 3

Relation between attendance problems of directors and gender diversity in director-level data.

The sample consists of an unbalanced panel of director data from 1,939 firms for the period 1996–2003, which were both in the Investor Responsibility Research Center (IRRC) Director Data and ExecuComp. We exclude inside directors from the sample, as well as all directors in a given fiscal year who were appointed that year. The dependent variable is a dummy variable that is equal to one if the director was named in the proxy as having attended less than 75% of meetings during the previous fiscal year. Compensation measures are quoted in 2003 dollars. Total compensation excludes meeting fees. Columns 1 and 3 show results from probit regressions; Columns 2 and 4 show results from linear probability (ordinary least squares) models. The specifications in Column 1 and 3 include two-digit standard industrial classification industry dummies. The specifications in Columns 2 and 4 include firm fixed effects. All specifications include year dummies. The sample is restricted to male directors in Columns 3 and 4. Standard errors are adjusted for potential heteroskedasticity and for group correlation at the directorship (director-firm) level in regressions without firm fixed effects. Absolute values of robust z-statistics are in brackets. Marginal effects for the female dummy and the fraction of women on the board are reported in parentheses in the probit regressions. The effect of the constant term is omitted. Asterisks indicate significance at 0.01 (***), 0.05 (**), and 0.10 (*) levels, and ^ indicates that coefficients are multiplied by 10,000. The number of observations may vary because of perfect predictability of the dependent variable.

Independent variable	Dependent variable: attendance problem			
	(1)	(2)	(3)	(4)
Female Dummy	−0.116*** (−0.007) [3.00]	−0.007*** [3.04]		
Fraction Female Directors			−0.417** (−0.026) [2.12]	−0.035* [1.71]
Meeting Fee	−0.043*** [2.66]	−0.001 [0.34]	−0.045*** [2.64]	−1.25^ [0.07]
# Other Directorships	0.055*** [6.00]	0.004*** [6.59]	0.061*** [6.14]	0.005*** [6.69]
Total Compensation	−1.05^ [1.38]	−0.001** [2.26]	−0.63^ [0.88]	−0.001* [1.89]
# Board Meetings	−0.020*** [4.63]	3.03^ [1.01]	−0.018*** [3.90]	0.001 [1.56]
Board Size	0.036*** [7.07]	0.002*** [2.69]	0.037*** [6.87]	0.002** [2.35]
Fraction Independent Directors	−0.189** [2.42]	0.020** [2.31]	−0.202** [2.44]	0.021** [2.29]
Tenure	−0.009*** [3.80]	−0.001*** [5.04]	−0.009*** [3.56]	−0.001*** [4.67]
Age	−0.002 [1.11]	−1.72^ [1.52]	−0.002 [0.73]	−1.57^ [1.27]
Retired Dummy	−0.129*** [3.87]	−0.006*** [3.44]	−0.132*** [3.79]	−0.006*** [3.17]
Log(Sales)	−0.043*** [3.80]	−0.003 [1.04]	−0.037*** [3.02]	−0.003 [1.04]
Tobin's q	−0.009 [1.20]	−3.64^ [0.79]	−0.014 [1.53]	−0.001 [1.59]
ROA	1.25^ [0.14]	0.04^ [0.05]	4.62^ [0.48]	0.47^ [0.60]
Volatility	0.138 [1.39]	−0.005 [0.42]	0.118 [1.13]	−0.006 [0.46]
Number of observations	63,998	65,480	56,951	58,302
R ²		0.07		0.08
Sample type	Full	Full	Men only	Men only
Industry effects	Yes	No	Yes	No
Firm fixed effects	No	Yes	No	Yes

(a) Table 3: Attendance and gender diversity

Table 4

Differentiating the effect of gender on attendance problems from the effect of new male directors.

The sample consists of an unbalanced panel of director data from 1,939 firms for the period 1996–2003, which were both in the Investor Responsibility Research Center (IRRC) Director Data and ExecuComp. We exclude inside directors from the sample. The dependent variable is a dummy variable that is equal to one if the director was named in the proxy as having attended less than 75% of meetings during the previous fiscal year. New Director is a dummy equal to one if the director was appointed in the current fiscal year. Fraction New Men is the ratio of nonexecutive male directors who were appointed in the current year to board size. Other sample characteristics are as in Tables 1 and 3. All columns show probit regressions. We exclude all directors in a given fiscal year who were appointed that year in Columns 2–4. The sample is further restricted to male directors in Columns 2 and 3. In Column 4 the sample is restricted to male directors with nonzero lagged Fraction of Years with Attendance Problems, which is defined to be the fraction of sample years a director was named as having attendance problems from his first appearance on a board up to and including the current fiscal year. All specifications include year and industry dummies. Standard errors are adjusted for heteroskedasticity and group correlation at the directorship (director-firm) level. Absolute values of robust z-statistics are in brackets. Asterisks indicate significance at 0.01 (***), 0.05 (**), and 0.10 (*) levels, and ^ indicates that coefficients are multiplied by 10,000. The number of observations may vary because of perfect predictability of the dependent variable.

Independent variable	Dependent variable: attendance problem			
	(1)	(2)	(3)	(4)
Fraction Female Directors		−0.426* [2.17]	−0.461** [2.18]	−0.996* [1.76]
Fraction New Men		−0.211 [1.12]		−0.19 [0.35]
Lagged Fraction New Men			−0.072 [0.38]	
Female Dummy	−0.113*** [2.93]			
New Director	−0.922*** [8.79]			
Meeting Fee	−0.042*** [2.61]	−0.046*** [2.66]	−0.054*** [2.86]	−0.081* [1.67]
# Other Directorships	0.055*** [5.97]	0.061*** [6.14]	0.058*** [5.43]	0.026 [1.05]
Total Compensation	−1.16^ [1.50]	−0.63^ [0.88]	−0.43^ [0.42]	−1.52^ [0.69]
# Board Meetings	−0.020*** [4.68]	−0.018*** [3.57]	−0.016*** [3.14]	−0.011 [0.73]
Board Size	0.035*** [6.95]	0.037*** [6.91]	0.037*** [5.79]	0.02 [1.37]
Fraction Independent Directors	−0.184*** [2.38]	−0.200*** [2.41]	−0.186*** [2.05]	−0.644*** [2.81]
Tenure	−0.009*** [3.79]	−0.009*** [3.56]	−0.009*** [3.21]	0.006 [1.00]
Age	−0.002 [1.02]	−0.002 [0.74]	−0.002 [0.88]	4.16^ [0.07]
Retired Dummy	−0.133*** [4.01]	−0.132*** [3.79]	−0.149*** [3.99]	−0.214*** [2.29]
Log(Sales)	−0.042*** [3.73]	−0.037*** [3.02]	−0.037*** [2.69]	−0.065* [1.80]
Tobin's q	−0.009 [1.18]	−0.012 [1.53]	−0.012 [1.09]	−0.053* [1.74]
ROA	1.87^ [0.21]	4.62^ [0.48]	0.003^ [1.76]	0.013*** [3.00]
Volatility	0.143 [1.46]	0.122 [1.16]	0.176 [1.48]	0.539* [1.93]
Number of observations	67,183	56,951	44,721	2,987
Sample type	Full	Men only, not newly appointed	Men only, not newly appointed	Men with past attendance problems, not newly appointed

(b) Table 4: New male director robustness

Table 5

The assignment of women to committees.

The sample consists of an unbalanced panel of director data from 1939 firms for the period 1998–2003, which were both in the Investor Responsibility Research Center (IRRC) Director Data and ExecuComp. We exclude inside directors from the sample. For each committee type we also restrict the sample to firms that have a committee of that type. The dependent variable in Column 1 is a dummy indicating whether a director is a member of the nominating, compensation, audit, or corporate governance committee in a given year. For each committee type, the dependent variable is a dummy variable indicating whether a director is a member of that committee in a given year. Fraction of Years with Attendance Problems is defined to be the fraction of sample years a director was named as having attendance problems from his first appearance on a board up to and including the current fiscal year. Other sample characteristics are as in Table 1. All columns show linear probability (ordinary least squares) models with firm fixed effects and year dummies. Standard errors are adjusted for potential heteroskedasticity. Absolute values of robust *t*-statistics are in brackets. Asterisks indicate significance at 0.01 (***), 0.05 (**), and 0.10 (*) levels, and “-” indicates that coefficients are multiplied by 10.000. The number of observations may vary because of perfect predictability of the dependent variable.

Independent variable	Dependent variable				
	Committee member (1)	Audit committee member (2)	Compensation committee member (3)	Nominating committee member (4)	Corporate governance member (5)
Female Dummy	0.035*** [7.34]	0.052*** [7.19]	-0.033*** [4.79]	0.020*** [2.72]	0.051*** [5.38]
Fraction of Years with Attendance Problems	-0.054*** [3.42]	-0.121*** [6.38]	-0.009 [0.46]	0.012 [0.52]	0.032 [1.02]
# Other Directorships	0.010*** [8.45]	-0.011*** [6.09]	0.026*** [14.49]	0.025*** [12.66]	0.026*** [10.57]
# Board Meetings	-3.33* [0.43]	-0.001 [0.55]	-0.001 [0.46]	-2.18* [0.17]	-0.001 [0.38]
Board Size	-0.019*** [12.39]	-0.020*** [9.65]	-0.020*** [9.83]	-0.015*** [6.25]	-0.015*** [4.80]
Fraction Independent Directors	0.025 [1.11]	-0.052* [1.74]	-0.032 [1.09]	0.002 [0.06]	-0.042 [0.80]
Tenure	0.001*** [4.52]	-0.002*** [5.54]	0.002*** [5.81]	0.007*** [18.20]	0.005*** [9.21]
Age	0.005*** [17.06]	0.001*** [4.25]	0.005*** [13.27]	0.004*** [10.21]	0.005*** [9.95]
Retired Dummy	-0.033*** [7.90]	-0.007 [1.39]	-0.010* [1.76]	-0.020*** [3.44]	-0.024*** [2.85]
Log(Sales)	0.001 [0.11]	-0.01 [1.00]	-0.007 [0.75]	-0.009 [0.71]	-0.035** [1.59]
Tobin's <i>q</i>	0.05 [0.05]	-0.001 [0.32]	0.002 [0.93]	0.004 [1.09]	0.000 [0.75]
ROA	-1.84* [1.19]	-0.78* [0.31]	-0.58* [0.24]	1.96* [0.60]	4.70* [1.07]
Volatility	-0.044 [1.49]	-0.012 [0.28]	-0.038 [0.95]	0.029 [0.61]	-0.053 [0.66]
Number of observations	54,458	54,267	54,099	41,508	24,853
<i>R</i> ²	0.15	0.08	0.09	0.12	0.12

- The fraction female coefficient is positive for total director compensation and fraction equity pay in key specifications.
- Table 8 finds no strong or robust relation between gender diversity and total CEO compensation or CEO incentive pay.
- This is consistent with women being underrepresented on compensation committees.

Economic message: Gender diversity affects director incentives more than CEO pay design. Diverse boards appear to change how directors are monitored and incentivized.

Table 6

Firm fixed effects regressions of chief executive officer (CEO) turnover on gender diversity.

The sample consists of an unbalanced panel of firm-level data from 1939 firms for the period 1996–2003, which were both in the Investor Responsibility Research Center (IRRC) Director Data and ExecuComp. CEO Turnover is a dummy equal to one if the CEO leaves the CEO position the following year. CEO turnover is defined to be missing in 2003. The sample is further restricted to turnover events that are not classified by ExecuComp as turnover due to CEO death. Stock Performance is the firm's raw return for the year net of the Center for Research in Security Prices value-weighted index, both compounded continuously. CEO Gender is a dummy that is one if the CEO is female. CEO Chairman is a dummy that is one if the CEO is also chairman. CEO Stockholdings are measured in percent. The source for CEO Age, CEO Stockholdings, CEO Gender, and CEO Chairman status is IRRC. The source for CEO Tenure is ExecuComp. Remaining sample characteristics are as in Table 1. All columns report linear probability models with firm fixed effects and year dummies. Standard errors are adjusted for potential heteroskedasticity. Absolute values of *t*-statistics are in brackets. Asterisks indicate significance at 0.01 (***), 0.05 (**), and 0.10 (*) levels, and “-” indicates that coefficients are multiplied by 10.000.

Independent variable	Dependent variable: CEO Turnover		
	(1)	(2)	(3)
Fraction Female Directors	-0.033 [0.26]	-0.042 [0.34]	-0.041 [0.34]
Fraction Female Directors times Stock Performance		-0.263* [2.41]	-0.252** [2.29]
Fraction Independent Directors times Stock Performance			-0.040 [0.75]
Stock Performance	-0.046*** [4.64]	-0.028** [2.19]	-0.004 [0.10]
CEO Age	0.022*** [11.03]	0.022*** [11.06]	0.022*** [11.05]
CEO Stockholdings	0.004** [2.15]	0.004** [2.14]	0.004** [2.14]
CEO Gender	-0.068 [0.60]	-0.069 [0.61]	-0.069 [0.61]
CEO Chairman	0.060** [3.04]	0.060** [3.05]	0.059** [3.04]
CEO Tenure	-3.94* [0.19]	-4.05* [0.19]	-4.12* [0.20]
Board Size	0.009* [1.80]	0.009* [1.86]	0.009* [1.86]
Fraction Independent Directors	-0.20*** [3.58]	-0.20*** [3.58]	-0.20*** [3.62]
Log(Sales)	0.012 [0.72]	0.012 [0.67]	0.012 [0.69]
# Business Segments	0.001 [0.51]	0.001 [0.50]	0.001 [0.51]
Volatility	-0.051 [0.70]	-0.056 [0.70]	-0.053 [0.71]
Number of observations	5,774	5,774	5,774
<i>R</i> ²	0.35	0.35	0.35

Table 8

Ordinary least squares and firm fixed effects regressions of measures of chief executive officer pay on gender diversity.

The sample consists of an unbalanced panel of firm level data from 1939 firms for the period 1996–2003, which were both in the Investor Responsibility Research Center (IRRC) Director Data and ExecuComp. Fraction CEO Incentive Pay is $1 - (\text{Salary} + \text{Bonus}) / \text{Total CEO Compensation}$. Total CEO Compensation is from ExecuComp and is the sum of Salary, Bonus, Other Annual, Total Value of Restricted Stock Granted, Total Value of Stock Options Granted (using Black and Scholes), Long-Term Incentive Payouts, and All Other Total. CEO Gender is a dummy that is one if the CEO is female. CEO Chairman is a dummy that is one if the CEO is also chairman. Data on CEO Age, CEO gender, and CEO chairman status are from IRRC. CEO Tenure is from ExecuComp. Remaining sample characteristics are as in Table 1. The dependent variable in Columns 1 and 2 is $\ln(\text{Fraction CEO Incentive Pay} + \epsilon)$, where ϵ is a very small number. The dependent variable in Columns 3 and 4 is $\ln(\text{Total CEO Compensation})$. The specifications in Columns 2 and 4 include firm fixed effects. The specifications without firm fixed effects include two-digit standard industrial classification industry dummies. All specifications include year dummies. Standard errors are adjusted for potential heteroskedasticity and for group correlation at the firm level in regressions without firm fixed effects. Absolute values of robust *t*-statistics are in brackets. Asterisks indicate significance at 0.01 (***), 0.05 (**), and 0.10 (*) levels, and “-” indicates that coefficients are multiplied by 10.000.

Independent variable	Dependent variable			
	Logtransform(Fraction CEO Incentive Pay)		Ln(Total CEO Compensation)	
	(1)	(2)	(3)	(4)
Fraction Female Directors	-0.156 [0.45]	0.779 [1.47]	0.188 [0.88]	0.271 [0.94]
CEO Age	-0.015*** [3.21]	-0.002 [0.41]	-0.004 [1.59]	-0.001 [0.22]
CEO Gender	-0.112 [0.42]	-0.295 [0.99]	0.004 [0.29]	0.004 [0.63]
CEO Chairman	0.258*** [4.23]	0.163*** [1.97]	0.163*** [4.56]	0.033 [0.98]
CEO Tenure	-0.027*** [4.99]	-0.037*** [5.48]	-0.005 [1.64]	-0.005 [1.83]
Board Size	0.018 [1.36]	0.008 [0.47]	0.016** [2.01]	0.001 [0.13]
Fraction Independent Directors	0.139*** [3.54]	-0.21 [0.88]	0.238** [2.33]	-0.219 [1.59]
Log(Sales)	0.295*** [12.08]	0.105 [1.15]	0.443*** [26.28]	0.378*** [9.17]
# Business Segments	-0.01 [1.38]	-0.022*** [2.68]	-0.006 [1.57]	-0.011*** [2.88]
Tobin's <i>q</i>	0.113*** [6.17]	0.035* [1.90]	0.089*** [3.27]	0.045*** [3.27]
ROA	-0.004*** [0.63]	-0.002 [1.24]	-2.23* [0.20]	0.001 [1.06]
Volatility	1.560*** [7.34]	0.009 [0.03]	0.731*** [5.83]	-0.085 [0.43]
Number of observations	7,411	7,542	7,605	7,697
<i>R</i> ²	0.17	0.55	0.41	0.72
Industry dummies	Yes	No	Yes	No
Firm fixed effects	No	Yes	No	Yes

(a) Table 7: Director pay

(b) Table 8: CEO pay

5.7 Tables 9 and 10: Performance effects

Read:

- Table 9 studies Tobin's *q*. OLS suggests a weak positive relation, but firm fixed effects and IV estimates show negative effects.

Table 9
Performance: Ln(Tobin's q) and gender diversity.
The sample consists of an unbalanced panel of firm-level data from 1,939 firms for the period 1996–2003, which were both in the Investor Responsibility Research Center (IRRC) Director Data and ExecuComp. Tobin's q is the ratio of the firm's market value to its book value of assets. Market value is book assets + book equity + market value of equity. Remaining sample characteristics are as in Table 1. The dependent variable in Columns 1–2 and 4–5 is the natural logarithm of Tobin's q. The specification in Column 1 includes two-digit standard industrial classification industry dummies. The specifications in Columns 2–4 include firm fixed effects. Column 3 reports the first stage of an instrumental variables (IV) regression with Fraction Males with Board Connections to Female Directors as an instrument for Fraction Female Directors. Column 4 reports the results of the IV estimation. Column 5 reports the results of an Arellano and Bond one step regression. All specifications include year dummies. Standard errors are adjusted for potential heteroskedasticity in Columns 2–5. Standard errors are adjusted for group correlation at the firm level in Column 1. Absolute values of t-statistics or z-statistics are in brackets. Asterisks indicate significance at 0.01 (***), 0.05 (**), and 0.10 (*) levels, and ° indicates that coefficients are multiplied by 10,000.

Independent variable	Dependent variable				
	Ln(Tobin's q)		Fraction Female Directors	Ln(Tobin's q)	
	(1)	(2)		(4)	(5)
Fraction Female Directors	0.221* [1.72]	-0.135* [1.67]		-5.924** [2.21]	-1.895 [0.56]
Board Size	-0.012*** [2.81]	-0.012*** [4.14]	3.59° [0.89]	-0.010** [2.56]	-0.103 [0.89]
Fraction Independent Directors	-0.131** [2.29]	0.04 [1.11]	0.051*** [9.89]	0.342*** [2.32]	-3.471** [2.20]
Log(Sales)	0.031*** [3.11]	-0.002 [0.17]	0.002 [0.20]	0.01 [0.72]	-0.553** [1.96]
# Business Segments	-0.016*** [7.05]	-0.003** [2.54]	0.63 [0.35]	-0.003* [1.67]	0.098*** [2.91]
Fraction Males with Board Connections to Female Directors			0.015*** [3.39]		
Lagged Ln(Tobin's q)					0.480*** [5.50]
Number of observations	9,209	9,477	9,477	9,477	5,588
R ²	0.25	0.11	0.08		
Industry dummies	Yes	No	No	No	No
Firm fixed effects	No	Yes	Yes	Yes	Yes
Regression type	OLS	Fixed effects	First-stage IV with fixed effects	IV with fixed effects	Arellano and Bond one step

Table 10
Performance: Return on assets (ROA) and gender diversity.
The sample consists of an unbalanced panel of firm-level data from 1,939 firms for the period 1996–2003, which were both in the Investor Responsibility Research Center (IRRC) Director Data and ExecuComp. ROA is net income before extraordinary items and discontinued operations divided by book assets. Remaining sample characteristics are as in Table 1. The dependent variable in all columns is ROA. The specification in Column 1 includes two-digit standard industrial classification industry dummies. The specifications in Columns 2 and 3 include firm fixed effects. Column 3 reports the results of an instrumental variables (IV) estimation with Fraction Males with Board Connections to Female Directors as an instrument for Fraction Female Directors. The first stage of the IV estimation is the same as in Column 3 of Table 9. Column 4 reports the results of an Arellano and Bond one step regression. All specifications include year dummies. Standard errors are adjusted for potential heteroskedasticity in Columns 2–4. Standard errors are adjusted for group correlation at the firm level in Column 1. Absolute values of t-statistics or z-statistics are in brackets. Asterisks indicate significance at 0.01 (***), 0.05 (**), and 0.10 (*) levels.

Independent variable	Dependent variable: ROA			
	(1)	(2)	(3)	(4)
	(1)	(2)	(3)	(4)
Fraction Female Directors	6.190* [1.89]	–6.170* [1.71]	–231.409** [2.07]	–159.658 [0.81]
Board Size	–0.327*** [2.84]	–0.276** [2.20]	–0.189 [1.18]	4.552 [0.67]
Fraction Independent Directors	–3.787*** [2.82]	1.997 [1.26]	13.719** [2.24]	30.618 [0.28]
Log(Sales)	2.716*** [6.09]	4.053*** [8.31]	4.603*** [6.97]	–25.254 [0.86]
# Business Segments	–0.03 [0.58]	–0.063 [1.10]	–0.049 [0.70]	2.332 [0.66]
Lagged ROA				0.271** [2.55]
Number of observations	9,324	9,553	9,553	5,656
R ²	0.07	0.03		
Industry dummies	Yes	No	No	No
Firm fixed effects	No	Yes	Yes	Yes
Regression type	Ordinary least squares	Fixed effects	IV with fixed effects	Arellano–Bond one step

- Table 10 studies ROA. The same pattern appears: naive estimates can be positive, but fixed-effect and IV estimates are negative.
- The Arellano–Bond specifications are less consistently significant, but they do not support a simple positive diversity-performance relation.
- These results show why raw correlations in the popular press can be misleading.

Economic message: More monitoring is not always better. Gender diversity may lower performance when it leads to excessive monitoring or excessive board intervention.

5.8 Table 11: Governance interaction

Read:

- The G-index measures antitakeover provisions and weak shareholder rights.
- Fraction Female Directors is negative in key performance specifications.
- The interaction between G-index and Fraction Female Directors is positive.
- This means female board representation is more valuable in weak-governance firms.
- In strong-governance firms, additional monitoring can be detrimental.

Economic message: The value of gender diversity depends on the governance environment. Female directors can improve value where monitoring is lacking, but quotas can harm firms that are already well governed.

Table 11

Performance and interaction of gender diversity with the Investor Responsibility Research Center (IRRC) shareholder rights index. The sample consists of an unbalanced panel of firm-level data from 1939 firms for the period 1998–2003, which were both in the IRRC Director Data and ExecuComp. Tobin's q is the ratio of the firm's market value to its book value of assets. Market value is book assets + book equity + market value of equity. ROA is net income before extraordinary items and discontinued operations divided by book assets. Gindex is the governance index from Compustat, Ibbotson, and Metrick (2003). This index is the sum of 24 dummy variables indicating a firm has a provision making it more difficult to be taken over. Data on governance provisions are from the IRRC. Remaining sample characteristics are as in Table 1. The dependent variable in Columns 1, 3, and 5 is Ln(Tobin's q). The dependent variable in Columns 2, 4, and 6 is ROA. The specifications in Columns 1–4 include firm fixed effects. Columns 5 and 6 report the results of Arellano and Bond one step regressions. All specifications include year dummies. Standard errors in all columns are adjusted for potential heteroskedasticity. Absolute values of robust t -statistics or z -statistics are in brackets. Asterisks indicate significance at 0.01 (*), 0.05 (**), and 0.10 (***) levels.

Independent variable	Dependent variable					
	Ln(Tobin's q) (1)	ROA (2)	Ln(Tobin's q) (3)	ROA (4)	Ln(Tobin's q) (5)	ROA (6)
Fraction Female Directors	−0.103 [0.90]	−5.036 [1.42]	−0.616* [1.87]	−22.500* [1.92]	−26.472** [2.22]	1,763.274 [0.83]
Gindex times Fraction Female Directors			0.055* [1.69]	1.857* [1.78]	3.194* [2.11]	−254.522 [0.97]
Gindex	−0.015** [2.17]	−0.411* [1.70]	−0.019** [2.53]	−0.561** [2.09]	0.028 [0.10]	21.625 [0.94]
Board Size	−0.005 [1.56]	−0.243* [1.91]	−0.004 [1.46]	−0.233* [1.83]	−0.189 [1.00]	10.398 [0.81]
Fraction Independent Directors	0.013 [0.28]	3.517** [2.18]	0.012 [0.27]	3.502** [2.17]	−3.970** [2.15]	−96.215 [0.41]
Log(Sales)	0.02 [1.08]	4.740*** [4.55]	0.02 [1.11]	4.759*** [4.57]	−0.419 [1.12]	−43.412 [0.98]
# Business Segments	−0.002 [1.20]	−0.023 [0.62]	−0.002 [1.24]	−0.025 [0.67]	0.080* [1.8]	3.840 [0.83]
Lagged Ln(Tobin's q)					0.299** [2.31]	
Lagged ROA						0.474 [1.17]
Number of observations	7384	7642	7384	7642	4,508	4,556
R^2	0.10	0.06	0.10	0.06		
Regression type	Firm fixed effects	Firm fixed effects	Firm fixed effects	Firm fixed effects	Arellano-Bond one step	Arellano-Bond one step

6 Short summary

- Female directors have better attendance records than male directors.
- Male directors have fewer attendance problems on more gender-diverse boards.
- Women are more likely to serve on audit, nominating, and governance committees.
- Gender-diverse boards make CEO turnover more sensitive to poor stock performance.
- Directors on gender-diverse boards receive more equity-based compensation.
- Gender diversity has no robust relation with CEO pay.
- The average relation between board gender diversity and performance is not positive after addressing endogeneity.
- In weak-governance firms, gender diversity can improve performance.
- In strong-governance firms, gender diversity can reduce performance, consistent with overmonitoring.
- The paper supports the idea that female directors affect governance, but it does not support mandatory quotas as a universal value-enhancing policy.

7 Conclusion

7.1 Core conclusion

Adams and Ferreira show that women in the boardroom have substantial effects on board behaviour. Female directors attend more, influence male attendance, and are placed on important monitoring committees. Gender-diverse boards also hold CEOs more accountable for poor performance.

7.2 Economic intuition

Boards face a tradeoff between advising and monitoring. Female directors appear to increase monitoring intensity. When a firm is poorly governed, this additional monitoring can improve value. When a firm is already well governed, additional monitoring can become excessive and reduce value by interfering with managers or weakening the advisory role of the board.

7.3 Takeaway

The paper's central lesson is that diversity changes governance, but governance is not mechanically value-enhancing. The effect of women directors depends on the firm's existing governance structure. This makes gender quotas potentially costly for well-governed firms, even though gender diversity may be valuable for firms that need stronger monitoring.